9 **(a)** In a circuit, a 4.0 V battery with internal resistance is connected in series to a thermistor and a coaxial resistor.

The coaxial resistor has a cylindrical core of length 0.020 m. The core is made of material A and is surrounded by material B as shown in Fig. 9.1. In the circuit, the direction of the current passing through the coaxial resistor is parallel to its length. The physical dimensions of the coaxial resistor are fixed.

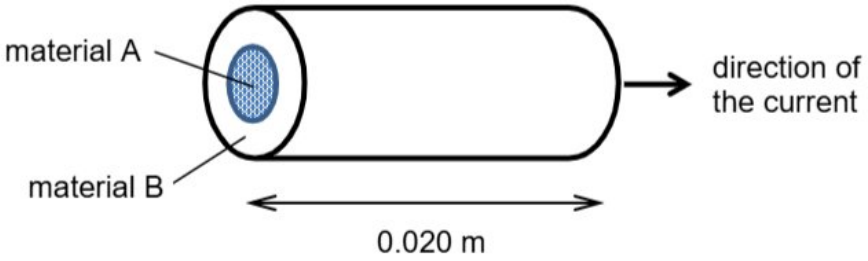


Fig. 9.1

(i) Draw a complete circuit diagram of the circuit in the space below.

Represent the coaxial resistor as two separate resistors A and B, one for each material respectively.

[3]

(ii) Fig. 9.2 shows the variation of the current I in the thermistor with temperature T of the thermistor.

Fig 9.3 shows the variation of I in the thermistor with voltage V across the thermistor.

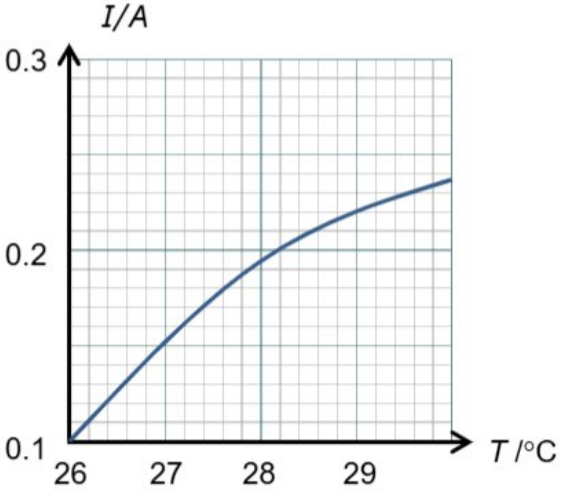


Fig. 9.2

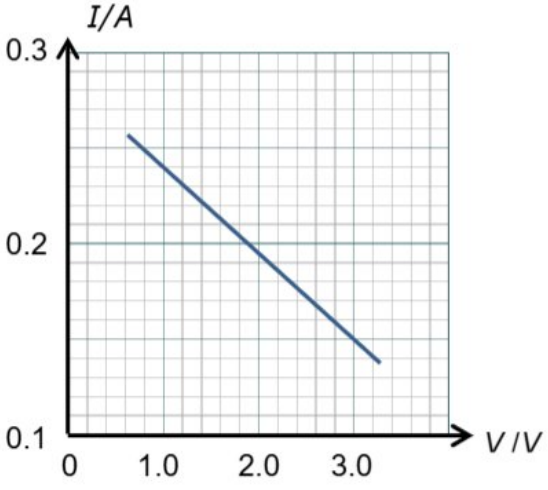


Fig. 9.3

Determine the resistance of the thermistor at 27 °C.

resistance = ... Ω [2]

(iii) Describe and explain what will happen to the potential difference across resistors A and B in the circuit when the temperature of the thermistor increases.

.....

.....

.....

.....[2]

(b) (i) Two identical lamps are connected to a d.c. and a.c. supply as shown in Fig. 9.4a and Fig. 9.4b respectively.

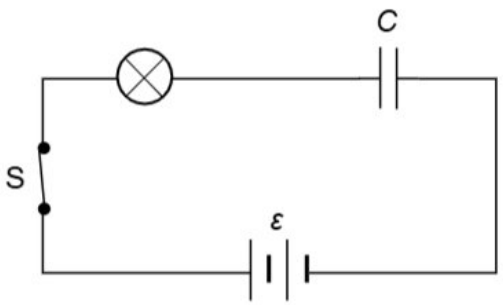


Fig 9.4a

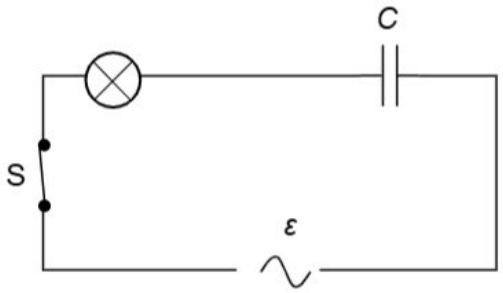


Fig 9.4b

State and explain which lamp will be turned on for a longer period of time. Assume that the time constant is large.

.....

.....

.....

.....

.....[3]

(ii) The charged capacitor of capacitance C is being discharged through a resistor of resistance R, as shown in Fig. 9.5. At time = 0 s, the capacitor is fully charged.

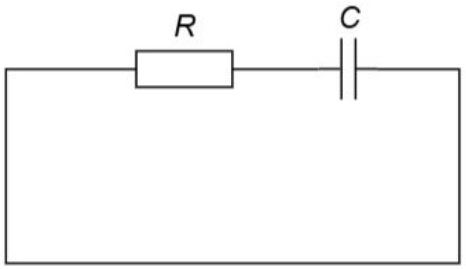


Fig. 9.5

The energy stored in the capacitor decreases with time as the capacitor discharges. Determine the time t, in terms of R and C, it takes for this stored energy to be discharged by 75% of its initial value.

t = ... [3]

(c) After a long time, the capacitor in Fig. 9.5 is fully discharged and a battery of e.m.f E of negligible internal resistance is used to recharge the capacitor in a circuit as shown in Fig. 9.6.

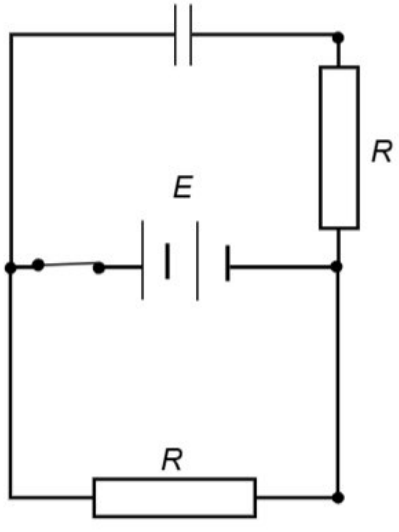


Fig. 9.6

(i) Determine the current, in terms of E and R, in the battery immediately after the switch is closed.

current = ... [2]

(ii) Determine the current, in terms of E and R, in the battery a long time after the switch is closed.

current = ... [2]

(iii) Hence suggest an expression for the time constant, in terms of R and C.

Explain your reasoning.

.....

.....

.....

.....

.....

.....[3]

[Total: 20]